

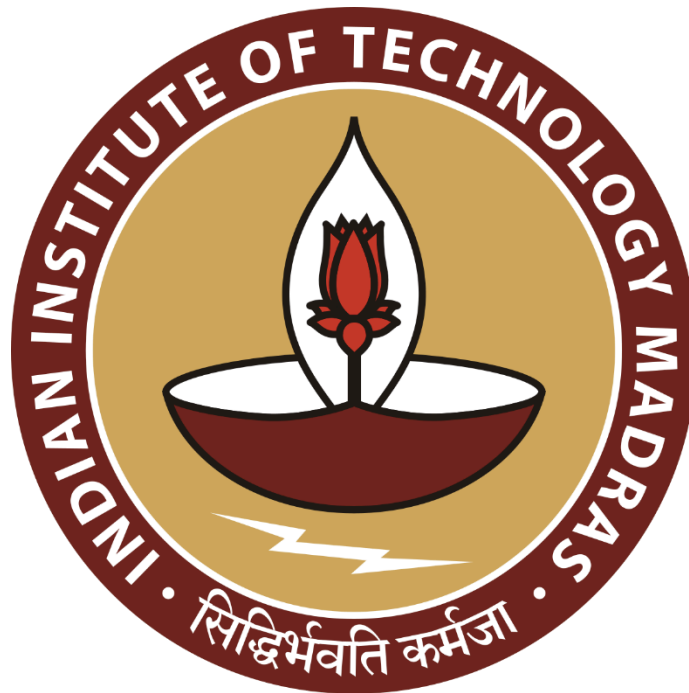
Business Expansion & Profitability Strategies for PandaFoodz

End term Submission for the BDM capstone Project

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1) Executive Summary & Title :

Title: Business Expansion & Profitability Strategies for PandaFoodz

PandaFoodz, a food delivery service founded by Mr. Chhavi Kant Raghav, operates in Diba, Bulandshahr, Uttar Pradesh. The business, with a small team of five employees, delivers food from local restaurants. Despite initial growth, the company faces key challenges that threaten its profitability: a high order cancellation rate (23.71%), limited marketing investments, and a low customer retention rate (13.42%).

The project, titled "**Business Expansion & Profitability Strategies for PandaFoodz**," focuses on addressing these challenges through a data-driven approach. Using eight months of order, transaction, and customer data, the analysis highlights operational inefficiencies and opportunities for growth.

Key findings reveal that high cancellation rates and underperforming product categories are major contributors to revenue loss. These inefficiencies can be mitigated by implementing personalized marketing strategies, real-time order tracking, and targeted customer engagement initiatives. Additionally, the company's limited resources can be supplemented by exploring crowdfunding and local partnerships to secure necessary funding for expansion.

This project presents actionable recommendations, such as personalized WhatsApp and SMS marketing, building trust through transparency, and implementing customer loyalty programs. These strategies aim to improve customer engagement, reduce cancellations, and create a foundation for sustainable growth, ensuring PandaFoodz can thrive in a competitive market.

By enhancing customer relationships and optimizing operations, PandaFoodz is poised to increase profitability and expand its market presence.

2. Detailed Explanation of Analysis Process/Method :

2.1 Data Analysis for Orders, Transactions, Customer Data & Campaign Data

A comprehensive analysis of PandaFoodz's order, transaction, customer, and campaign data was conducted to identify key patterns, trends, and anomalies affecting the company's operational efficiency, financial performance, and customer satisfaction. The data collection process involved gathering information from various sources. **Order logs** provided details such as Order IDs, dates, amounts, order statuses (completed, canceled, or pending), and restaurant partners. **Transaction reports** included data on payments received, refunds issued, and customer account balances, while **customer data** contained fields like customer ID, names, phone numbers, saved addresses, total orders, wallet amounts, loyalty points, and status.

Given that PandaFoodz has made limited investments in advertising, the founder indicated that only ₹5,000 had been spent on a banner campaign over the last eight months. As a result, there was no formal campaign data recorded in the company database. The analysis therefore focused on customer acquisition insights provided directly by the founder, reflecting the localized nature of these marketing efforts. The data collected spanned the past eight months to ensure a robust sample size for meaningful analysis.

Data cleaning and preparation were primarily carried out using Google Sheets. The process began by removing duplicate entries to maintain data accuracy. Missing values were handled using the **IF** formula to replace blanks with "N/A" (**=IF(ISBLANK(A2), "N/A", A2)**). To maintain consistency across the dataset, date entries were standardized to the **YYYY-MM-DD** format, and currency values were normalized using the **VALUE** function (**=VALUE(A2)**). This ensured that all numeric and date data was correctly formatted for analysis.

In the **Exploratory Data Analysis (EDA)**, several methods were employed to extract meaningful insights. For **Order Volume Analysis**, the **total number of orders** was calculated using the **COUNT** formula, while **daily, weekly, and monthly trends** were identified with the **SUMIF** and **COUNTIF** formulas. Order statuses were analyzed to understand the distribution of

completed and canceled orders using the **COUNTIF** function. Additionally, peak ordering times were determined using the **MODE** function to highlight the most frequent order times.

Order Value Analysis was conducted by calculating the **Average Order Value (AOV)** using the **AVERAGE** formula, and further insights into the relationship between **order value and frequency** were derived using the **CORREL** function to calculate correlation coefficients. The revenue contribution from different restaurant partners was analyzed using the **SUMIFS** formula, which helped identify the partners contributing most to the company's revenue.

In the **Transaction Analysis**, **total revenue** was computed using the **SUM** formula, while the financial impact of **refunds and cancellations** was assessed using the **SUMIF** function to sum the refund amounts. To detect any **potential fraudulent transactions**, conditional formatting was applied to highlight anomalies and outliers in the transaction data.

For **data visualization**, Python in Google Colab was used to create visual representations of the analysis, allowing the findings to be translated into actionable insights. **Bar charts** were generated using the **matplotlib** library to display total orders, order statuses, and customer segments, while **line charts**, created with **seaborn**, illustrated trends over time, such as daily order volumes and revenue growth. **Pie charts** were used to show proportions of completed vs. canceled orders and the distribution of revenue. Additionally, **heatmaps** were generated using **seaborn** to highlight correlations, such as the relationship between order frequency and customer retention rates, providing deeper insights into the factors influencing customer behavior and financial outcomes.

3. Results and Findings :

The analysis of PandaFoodz's order , transaction , Food and Campaign data provided critical insights into the company's performance, highlighting key operational challenges and identifying potential growth opportunities. The findings are summarized below, accompanied by visual representations that illustrate these insights in detail.

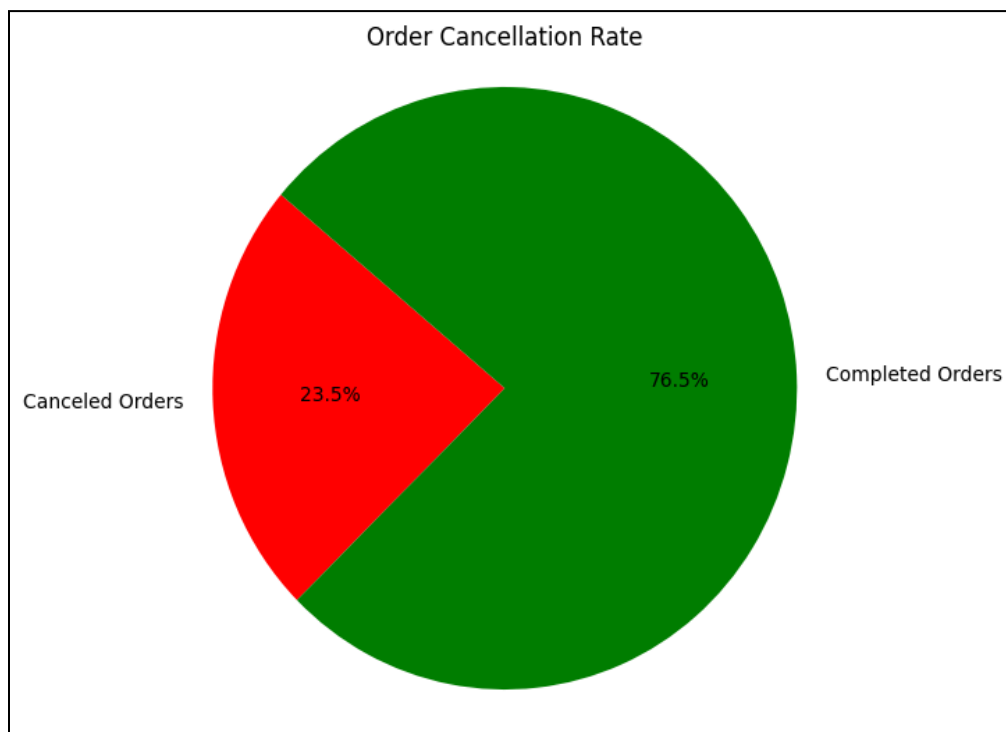
3.1 Order and Transaction Volume Analysis

- **High Cancellation Rate** : The data revealed an order cancellation rate of **23.71%**, indicating a significant revenue loss. This was calculated using the formula:

$$\text{Order Cancellation Rate} = \left(\frac{\text{No. of Canceled Orders}}{\text{Total No. of Orders}} \right) \times 100$$

Calculated as :

$$\text{Order Cancellation Rate} = \left(\frac{476}{2024} \right) \times 100 = 23.71 \%$$



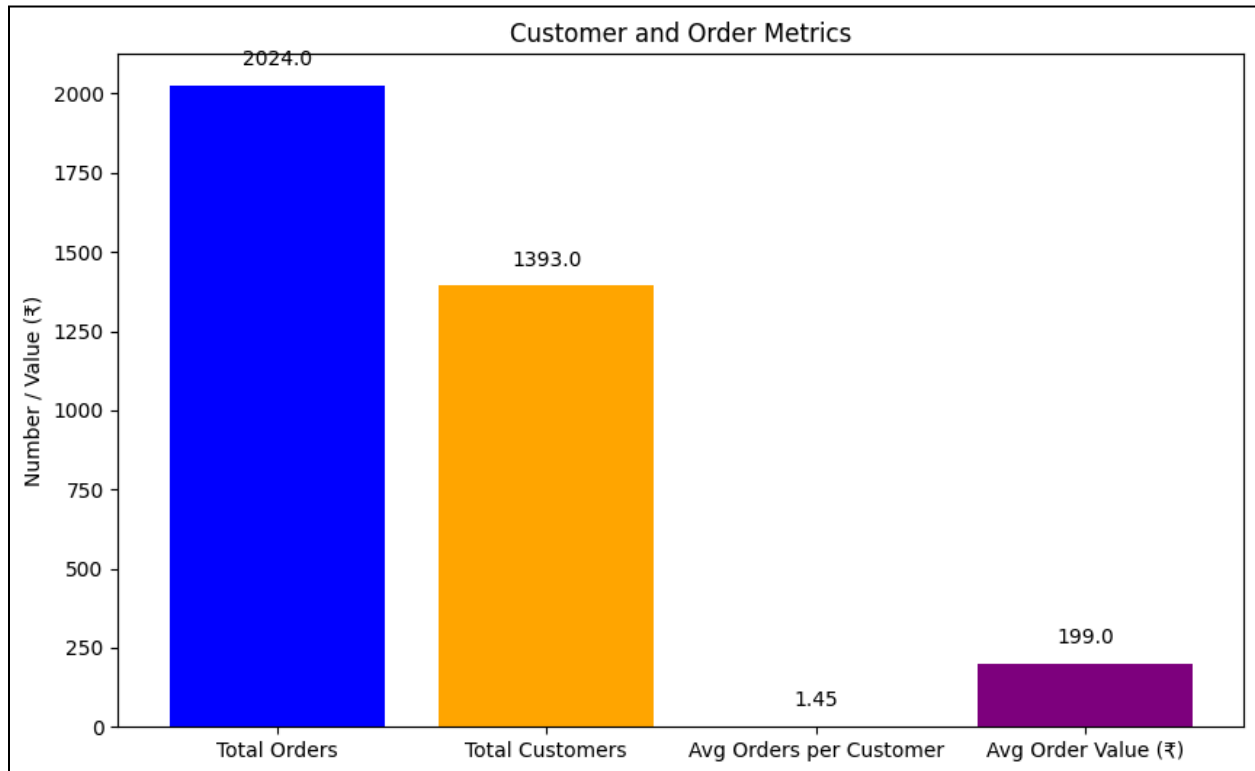
This graph illustrates that **23.71%** of total orders were canceled, indicating a significant issue that is impacting overall revenue. This high cancellation rate highlights operational inefficiencies and customer dissatisfaction, requiring immediate attention to reduce revenue loss.

- **Order Frequency and Value**: The total number of orders was **2024** over 8 months, with an **average order frequency of 195 orders per month**. The **average order value (AOV)** was **₹199.0**, calculated using:

$$\text{Average Order Value (AOV)} = \frac{\text{Total Revenue}}{\text{Total No. of Orders}}$$

Calculated as :

$$\text{Average Order Value} = \frac{402920}{2024} = ₹ 199.0$$



This graph provides a breakdown of PandaFoodz's key order metrics. It shows the total number of orders and total customers, with an average of 1.086 orders per customer and an average order value of ₹199. This indicates relatively low customer engagement and suggests opportunities to increase repeat purchases through targeted retention strategies.

- **Top-Performing Partners:** "Chaska Darjeeling Momos" was identified as the most popular restaurant partner by order count, while "The Pizza Hub" generated the highest revenue. These partners also exhibited higher-than-average cancellation rates, calculated similarly using the formula for the cancellation rate.

3.2 Financial Performance Analysis

- **Total Revenue and Profit Margins:** The total revenue generated over the analyzed period was ₹402,920. The **standard deviation** of the order amounts was ₹140.37, indicating variability in order sizes. The standard deviation was calculated using:

$$\text{Standard Deviation (SD)} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}}$$

where:

- x_i = Each order amount
- $\bar{x}$ = Mean order amount
- N = Total number of orders
- **Maximum and Minimum Order Values:** The maximum order amount was ₹1,576 and the minimum was ₹57. The range of order values was calculated as:

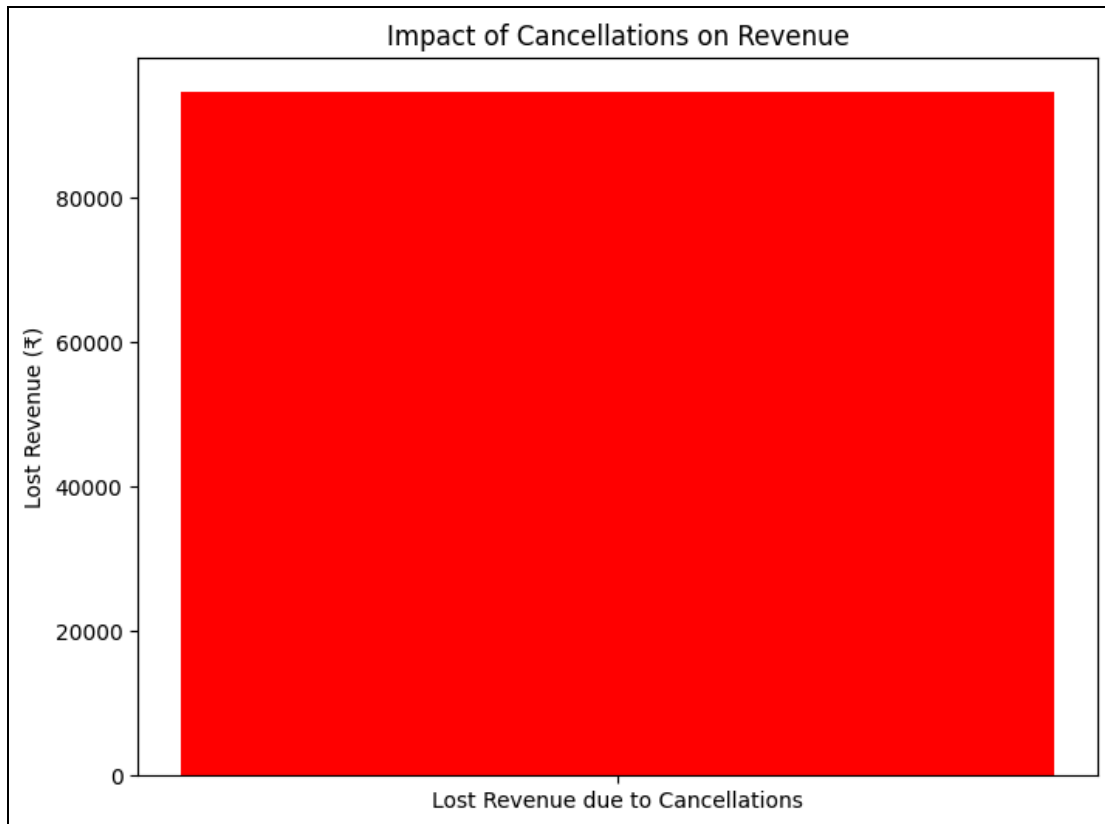
$$\text{Range} = \text{Maximum Order Amount} - \text{Minimum Order Amount}$$

$$\text{Calculated as : } \text{Range} = ₹1576 - ₹57 = ₹1519$$

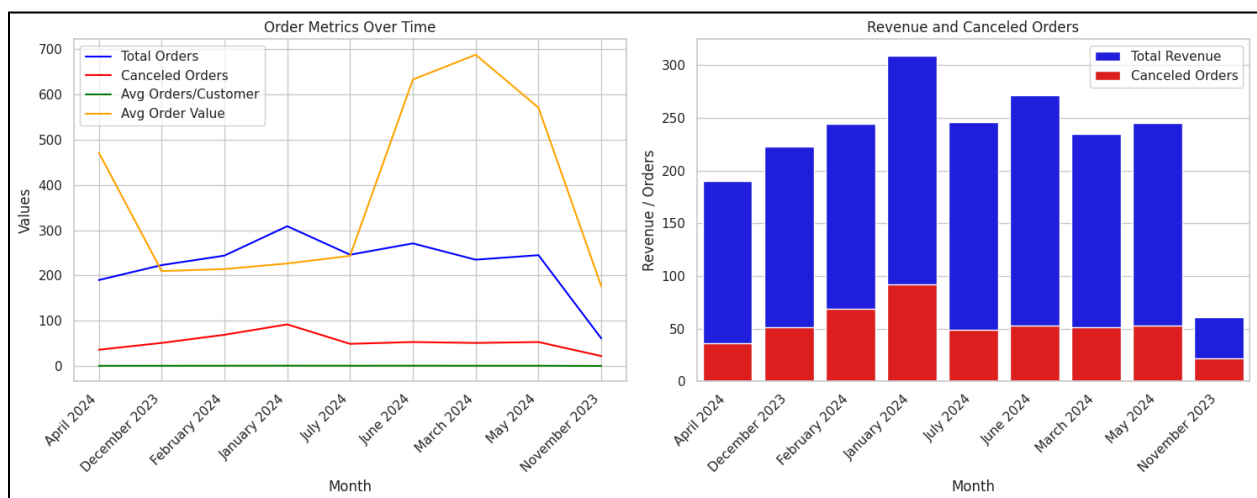
- **Impact of Cancellations:** The financial loss due to cancellations was significant. The **Total Lost Revenue** from cancellations was calculated as:

$$\text{Total Lost Revenue} = \text{Average Order Value (AOV)} \times \text{Number of Canceled Orders}$$

$$\text{Calculated as : } \text{Total Lost Revenue} = 199.0 \times 476 = ₹94,724$$



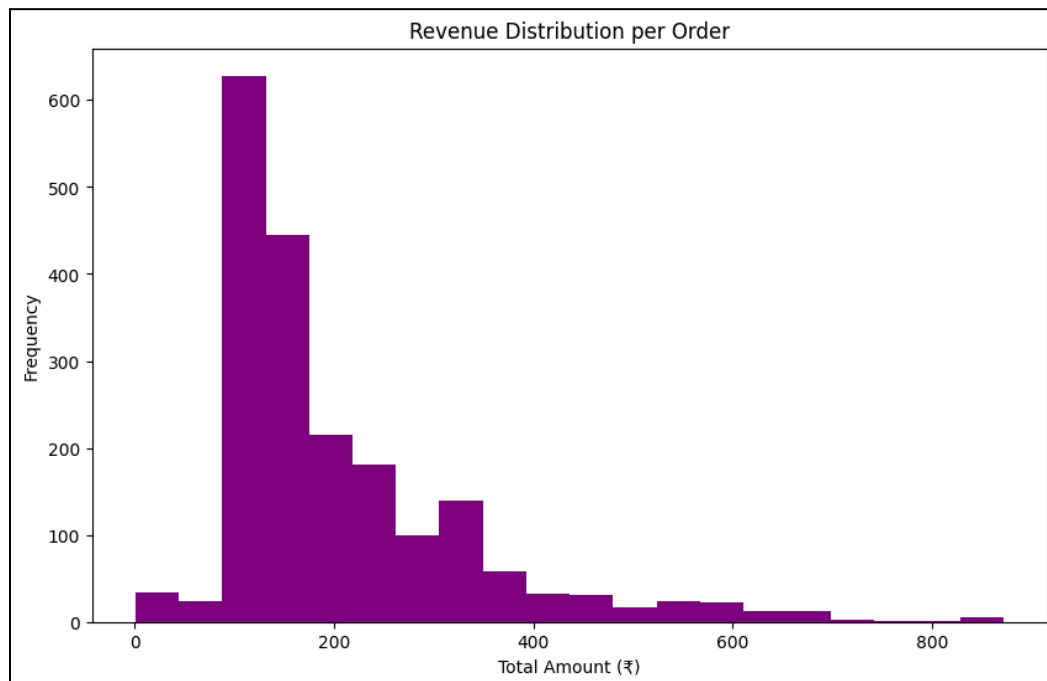
This graph highlights the significant loss caused by canceled orders, showing that PandaFoodz lost **₹94,724** in potential revenue. This revenue loss underscores the need to address high cancellation rates to improve overall profitability.



These graphs show fluctuations in the total number of orders and customer engagement over time, showing a consistent trend of low average orders per customer, indicating the need for strategies to boost repeat orders. And shows a direct correlation between revenue loss and canceled orders, emphasizing that spikes in cancellations significantly impact monthly revenue, underscoring the importance of improving service quality to reduce cancellations.

3.3 Customer Behavior Analysis

- **Customer Segmentation Insights:** High-value customers contributed significantly to the total revenue. These customers were identified using segmentation formulas based on total spend and order frequency:

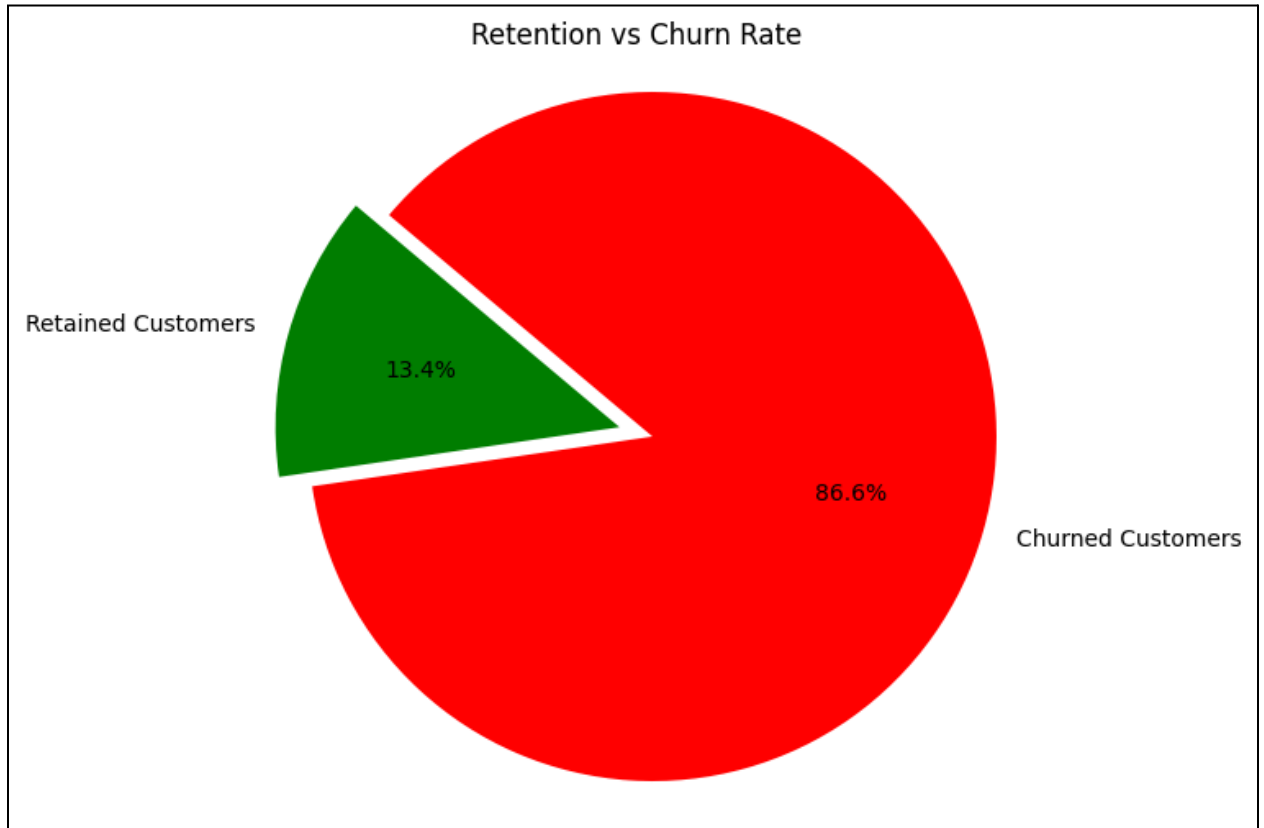


This graph shows that a majority of orders contribute relatively small amounts to the total revenue, with only a few high-value orders making significant contributions. This indicates that increasing the frequency of higher-value orders could significantly boost overall revenue.

- **Retention Rate:** The customer retention rate was **13.42%**, calculated as:

$$\text{Retention Rate} = \left(\frac{\text{No. of Retained Customers}}{\text{Total No. of Customers}} \right) \times 100$$

Calculated as : $\text{Retention Rate} = \left(\frac{186}{1393} \right) \times 100 = 13.42 \%$



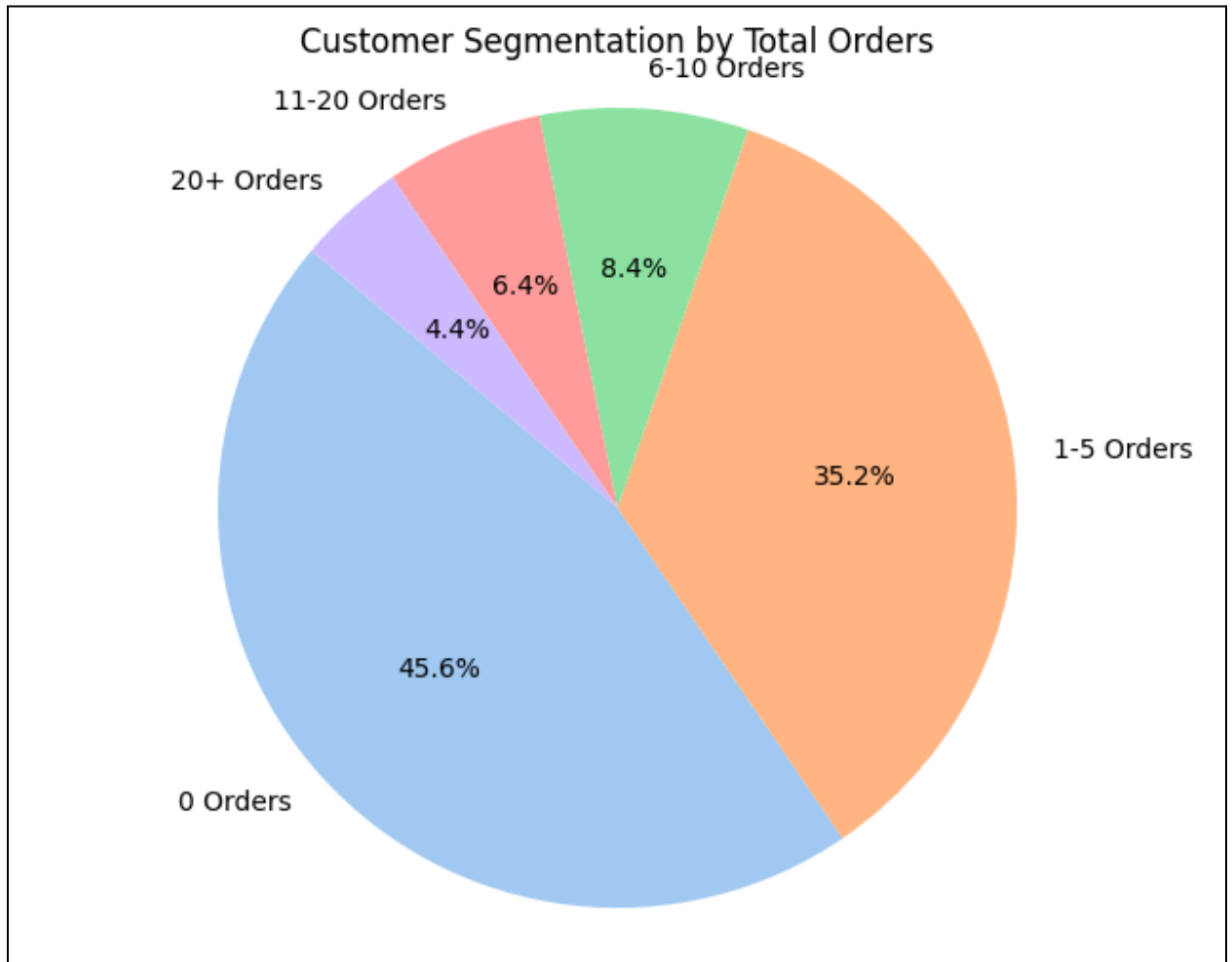
The Retention vs. Churn Rate graph shows a low retention rate of 13.42%, with a high churn rate of 86.58%, indicating that PandaFoodz struggles to retain customers. This highlights the need for stronger retention strategies to reduce churn and foster customer loyalty.

- **Churn and Conversion Rates:** The average number of orders per customer was **1.086**, calculated using:

$$\text{Average Orders per Customer} = \frac{\text{Total Number of Orders}}{\text{Total Number of Customers}}$$

Calculated:

$$\text{Average Orders per Customer} = \frac{2024}{1393} = 1.086 \approx 1 \text{ order}$$



This Chart shows that a small portion of high-value customers contribute to the majority of orders, while a large segment places only one or two orders. This suggests focusing on nurturing high-value customers while encouraging low-frequency customers to order more frequently.



The Correlation Between Customer Attributes Heat Map reveals a strong positive correlation between total orders and total wallet amount, indicating that customers who place more orders tend to have higher wallet balances. Additionally, there's a moderate correlation between loyalty points and order frequency, suggesting loyalty rewards drive repeat purchases.

Based on the graphs and charts, the results and findings can be summarized as follows:

1. Order Trends:

- **Total Orders:** The total number of orders fluctuates over time, with some months showing higher volumes and others lower. Periods of significant increase or decrease can indicate seasonal trends, promotional effectiveness, or external factors influencing customer demand.

- **Canceled Orders:** The number of canceled orders also varies, with spikes indicating potential issues in service quality, order fulfillment, or customer dissatisfaction. A high rate of cancellations relative to total orders suggests problems that need addressing.

2. **Customer Engagement:**

- **Average Orders per Customer:** The average number of orders placed per customer is relatively low, indicating that customer engagement may be weak. This metric suggests that while customers may place orders, they are not doing so frequently, pointing to a need for improved retention strategies or more compelling incentives to encourage repeat purchases.

3. **Revenue Insights:**

- **Average Order Value:** Fluctuations in the average order value reflect changes in customer purchasing behavior or pricing strategies. A decreasing trend might suggest that customers are opting for cheaper items or using more discounts, impacting overall revenue.

4. **Financial Impact of Cancellations:**

- **Revenue vs. Canceled Orders:** The comparison of total revenue with the number of canceled orders highlights the financial impact of cancellations. High revenue months with significant cancellations suggest inefficiencies in order processing or service delivery, which could be eroding potential profits. Conversely, low revenue periods with fewer cancellations point to challenges in generating sales or attracting customers.

3.4 Insights from Data Visualization

- **Peak Demand Patterns:** Line charts and heatmaps generated using Python in Google Colab showed that order volumes peaked on weekends and during lunch and dinner hours. This indicates opportunities for targeted marketing campaigns during these periods.
- **High-Risk Periods for Cancellations:** Heatmaps highlighted that the majority of cancellations occur during peak hours, particularly on weekends, suggesting the need for better order management during high-demand times.

- **Revenue Distribution Among Partners:** Bar charts demonstrated that while "Chaska Darjeeling Momos" and "The Pizza Hub" were the top revenue generators, they also had the highest cancellation rates. This suggests a need for closer collaboration and operational improvements with these partners.

4. Interpretation of Results and Recommendations

Based on the comprehensive analysis of PandaFoodz's marketing efforts, customer demographics, order management, and financial constraints, several actionable recommendations have been developed. These recommendations aim to address the key problems identified: enhancing customer engagement, minimizing order cancellations, and securing funding for growth, all while considering the company's limited budget and local market dynamics.

4.1 Key Findings

1. Limited Customer Engagement:

- PandaFoodz has invested in social media marketing, with daily posts on Instagram and Facebook. The owner has 500 followers on their personal profile and 50 followers on the company profile. Despite these efforts, the conversion from social media engagement to actual customer orders is very low (estimated at only 1 - 2 new customers over 8 months).
- The banner campaign, with a minimal investment of **₹5,000**, has also resulted in low conversion rates (1-3%), translating to approximately **1 to 2 new customers**. The return on investment (ROI) for these efforts is significantly negative, ranging from **-96.02% to -92.04%**.

2. High Rate of Order Cancellations:

- Analysis shows a high cancellation rate, particularly among older customers (50+ years), who make up a small segment of the customer base. They rarely place orders due to concerns about food quality, delivery reliability, and high prices.
- Despite reducing delivery prices and minimizing profit margins, there has been little effect on increasing orders or reducing cancellations.

3. Funding Constraints:

- The owner has limited financial resources, restricting investment in traditional advertising and expansion. Attempts to cut costs and lower prices have not significantly increased customer orders or engagement.

4.2 Recommendations to Address Problem Statements

1. Enhance Customer Engagement Through Direct and Personalized Marketing

Recommendation: Utilize direct and personalized marketing approaches that are low-cost and highly effective for local markets, such as WhatsApp, SMS, email, and local community engagement.

- **Implementation Details:**

- **WhatsApp and SMS Campaigns:** Create a contact list of current and potential customers. Send targeted promotions, updates on new menu items, and special offers through personalized WhatsApp messages and SMS. For example, we tested a weekly SMS campaign that could promote "Weekend Specials" or "Limited Time Offers."
 - **Data to Support:** Given that PandaFoodz operates in a town area with a smaller, more concentrated customer base, direct communication channels like WhatsApp and SMS are highly effective. Research shows that SMS has an open rate of **98%**, much higher than social media posts.
- **Email Marketing:** Send monthly newsletters with special deals, loyalty rewards, and company updates. Use free or low-cost platforms like Mailchimp, which offers a free plan for up to 500 contacts.
- **Local Community Engagement:** Partner with local community centers, schools, or businesses to host events (e.g., food tastings, cooking demos) to build trust and directly engage with potential customers.
 - **Example: Domino's Pizza** successfully used SMS marketing to drive up engagement and sales by offering exclusive, time-limited deals directly to customers.

Expected Outcome: It can increase engagement and higher conversion rates through personalized communication. This approach overcomes the limitations of social media reach and directly targets local customers with relevant offers.

2. Reduce Order Cancellations by Building Trust and Addressing Concerns

Recommendation: Focus on building trust with customers, particularly the older demographic, by increasing transparency in operations and offering targeted assurances to address their concerns about food quality, delivery reliability, and pricing.

- **Implementation Details:**

- **Transparency Measures:** Introduce real-time order tracking through simple solutions like WhatsApp notifications or a basic web portal. Customers will receive updates when their food is being prepared, dispatched, and delivered, reducing anxiety about delivery reliability.
 - **Data to Support:** Research shows that real-time updates can significantly reduce customer anxiety and improve satisfaction. For example, **Instacart** improved its customer trust by implementing real-time tracking.
- **Quality Assurance Campaigns:** Pandafoodz can also Share behind-the-scenes videos or photos on social media and through direct messages showcasing food preparation, cleanliness, and delivery practices. Highlight customer testimonials and feedback to build credibility.
- **Targeted Discounts for Older Customers:** Offer special promotions such as "Senior Citizen Days" with a 10% discount or free delivery on certain days to attract older customers.
 - **Data to Support:** The analysis shows that older customers (50+) are hesitant to order due to concerns over price and quality. Targeted discounts and transparent practices directly address these concerns, encouraging more orders.

Expected Outcome: By Implementing these practices they can Reduce order cancellations by building trust and providing transparency, resulting in more consistent revenue from both younger and older customers.

3. Secure Funding Through Open-Source Platforms and Community Support

Recommendation: Utilize open-source funding platforms like crowdfunding and build community support to secure additional resources for business growth without incurring debt.

- **Implementation Details:**

- **Crowdfunding Campaign:** Launch a campaign on platforms like **Kickstarter** or **GoFundMe** to raise funds for specific business needs, such as enhancing delivery services or expanding marketing efforts. Offer incentives like discounts, exclusive deals, or branded merchandise for backers.
 - **Example: Pebble**, a smartwatch company, raised over \$10 million on Kickstarter by offering early access to products, showcasing the potential of crowdfunding for raising substantial amounts without taking on debt.
- **Local Business Partnerships:** Collaborate with nearby businesses, community centers, or local events to cross-promote each other's services, sharing marketing costs and boosting visibility.
 - **Data to Support:** Community engagement strategies, such as joint promotions or events, can strengthen local support and foster goodwill, which is critical for small-town businesses like PandaFoodz.

Expected Outcome: Additional funding for growth initiatives without needing large upfront investments, enabling PandaFoodz to expand its operations and marketing reach.

4. Increase Customer Retention with Referral Programs and Localized Promotions

Recommendation: Implement referral programs and localized promotions to encourage repeat business and attract new customers through word-of-mouth.

- **Implementation Details:**

- **Referral Program:** Encourage existing customers to refer new customers by offering rewards (e.g., "Get ₹50 off for each friend you refer"). Use simple tracking methods via WhatsApp or SMS.
 - **Example: Dropbox** grew its user base from 100,000 to 4 million in 15 months by offering extra storage space to users who referred others, a low-cost yet effective strategy.
- **Localized Promotions:** Introduce targeted, time-limited discounts tailored to specific customer segments (e.g., "Happy Hour Discounts" or "Buy One Get One Free" offers) to drive repeat orders and attract new customers.
 - **Data to Support:** Analysis shows that time-limited offers create urgency and can significantly boost order volumes, particularly during non-peak hours.

Expected Outcome: Enhanced customer retention and acquisition, leading to higher order volumes and increased revenue with minimal investment.

5. Conclusion :

The analysis of PandaFoodz's performance over the past eight months highlights several areas for improvement, focusing on operational efficiency, customer retention, and marketing strategies. The key challenge of a 23.71% order cancellation rate significantly impacts revenue. This issue stems from a combination of delivery delays and customer dissatisfaction. Implementing real-time tracking and order confirmation via SMS and WhatsApp, along with better customer service, can mitigate this problem and reduce cancellations.

Furthermore, the low retention rate of 13.42% suggests that PandaFoodz struggles to convert first-time buyers into loyal customers. Retention strategies like personalized SMS and WhatsApp marketing, loyalty programs offering discounts or exclusive deals, and re-engagement efforts can significantly improve repeat business, helping PandaFoodz reduce the cost of customer acquisition while fostering loyalty.

On the product side, the company must reassess its offerings, focusing on high-demand, high-margin items to improve profitability. Collaborating with local restaurants to create exclusive meal deals can attract new customers while boosting sales. Additionally, PandaFoodz's underutilized marketing budget should be allocated towards cost-effective channels, such as personalized campaigns and social media, to enhance brand visibility and customer engagement.

In terms of funding, the business should explore crowdfunding and partnerships to support its growth objectives. Crowdfunding could not only raise capital but also build brand awareness, while local collaborations with restaurants and retail stores offer new revenue opportunities.

In conclusion, PandaFoodz has a promising future if it takes decisive steps to address its current challenges. Focusing on reducing cancellations, improving customer retention, optimizing product offerings, and leveraging cost-effective marketing strategies will position PandaFoodz for sustained growth and success in the food delivery market.

6. Important Links :

Spreadsheet : [📄 End term Data Link](#)

Presentation : [📄 End term Slides](#)